

Understanding the EDA Notebook

Phase 4 explained, one cell at a time

What is EDA, and why do it?

EDA stands for **Exploratory Data Analysis**. It is the stage where you get to know your data before you try to answer anything with it. You are not testing a theory yet. You are looking around, checking what is there, and finding the things that would trip you up later if you did not know about them.

A useful comparison: before a builder puts up a house, they survey the land. They are not building anything yet. They are finding out where the ground is soft, where water collects, where the rock is. Skip that step and the house still goes up — it just falls down later, and you never know why.

That is exactly what this notebook did. Nine cells. Each one asked a small question, and several of the answers changed how the rest of the project has to be done.

The nine cells at a glance

Cell	What it looked at	The one-line takeaway
1	Loading the data	Four tables opened correctly
2	What the dataset covers	One year, three months, 8 at-risk species
3	Survey effort	Effort is wildly uneven — the key finding
4	Species distributions	Some birds strongly prefer one habitat
5	Time patterns	Grassland birds go quiet as the morning warms
6	Weather	Heat reduces what you find; disturbance more so
7	Observers	People differ a lot — but harmlessly, here
8	How birds were detected	You cannot see through a forest
9	Summary	Everything gathered in one place

Cell 1 — Loading the data

The question we were asking	Can we open the four tables we built earlier, and are they the size we expect?
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What the code did

It opened four files. Think of them as four spreadsheets, each answering a different level of question. **rows** has one line per bird sighting. **sessions** has one line per survey visit. **species** has one line per bird species. **parks** has one line per national park.

It also loaded our own *stats_helpers* file — the maths code we wrote ourselves after Windows blocked the usual statistics library.

What we found

15,372 sightings, 1,408 survey sessions, 115 species profiled, 11 parks. All four opened without error.

An everyday way to think about it. It is like counting the pages of a document before you photocopy it. Quick, boring, and it saves you from discovering halfway through that something is missing.
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Why it matters

This is the equivalent of checking your tools before starting work. If a file had failed to open, or had the wrong number of rows, we would want to know now — not after building ten charts on top of it.

Cell 2 — What the dataset actually covers

The question we were asking

What time period, how many parks, how many birds? What are the boundaries of what we can say?

What the code did

It printed the basic dimensions of the data: the first and last survey date, how many parks and plots were visited, how many species were seen, and how many of those species are on the official at-risk list.

What we found

The surveys ran from **7 May to 19 July 2018** — 64 days, three months, **one year**. Across 11 parks and 609 plots, 3 observers recorded 15,372 sightings of 127 species. Of those, **378 sightings were at-risk species, but only 8 distinct species**.

An everyday way to think about it. Imagine measuring a child once. You can say how tall they are. You cannot say whether they are growing. One year of bird data has exactly that limitation.

Why it matters

Two limits come straight out of this.

One year means no trends. Anything we say about time has to be "within this one breeding season". We cannot say a population is rising or falling, because we only ever looked once.

378 sightings of 8 species is plenty to compare *how often* at-risk birds turn up in each habitat. It is not enough to say much about any individual species. The report should claim the first and stay quiet on the second.

Cell 3 — Survey effort — the most important cell

The question we were asking

How much surveying actually happened in each park and each habitat? Was it even?

What the code did

It counted survey sessions per park and habitat, then checked two more things: how many times each plot was revisited, and how long a session lasted.

What we found

The effort is **extremely uneven**. At Antietam (ANTI) there were **26 forest sessions and 252 grassland sessions** — almost ten times more. Seven parks have no grassland surveying at all. Grassland plots were revisited about 3 times each, forest plots only 2. But session *length* was a median of exactly **10 minutes in both habitats**.

An everyday way to think about it. Two shops, one open 12 hours a day, one open 1 hour. The first sells more. That tells you nothing about which shop is better — you would compare *sales per hour open*. "Species per session" is exactly the same idea.

Why it matters

This single cell is why the whole project is built the way it is.

If you simply counted sightings, grassland at Antietam would look far richer than forest — but only because somebody went there ten times more often. That is not biology, it is a scheduling record.

The 10-minute figure is the good news. Because a session means the same amount of effort in both habitats, we can fairly say "species *per session*". That becomes the unit for every comparison in the project.

Cell 4 — Which species dominate, and where

The question we were asking

Which birds are most common, and do forest and grassland hold different communities?

What the code did

It listed the twelve most-recorded species, then split each one by habitat, then counted how many species appear in only one habitat.

What we found

Northern Cardinal is the most common bird overall (1,125 sightings) and it is genuinely at home in both — 595 forest, 530 grassland. Others are lopsided: **Red-eyed Vireo** is 694 forest against 44 grassland; **Field Sparrow** is 36 forest against 456 grassland. Twenty species were recorded only in forest, nineteen only in grassland, and 88 in both.

An everyday way to think about it. If you only ever survey cats in cat cafes, you will conclude cats never go outdoors. The finding is about where you looked, not where the animal lives.

Why it matters

This is the raw material for the habitat-preference finding. Some birds clearly need one habitat; others do fine in either.

But there is a trap in the last number. "Twenty species only in forest" is misleading, because seven parks only ever had forest surveyed. A bird living in one of those parks would look forest-only even if it was perfectly happy in grassland — nobody ever checked. That is why our proper preference analysis only uses the four parks where *both* habitats were surveyed.

Cell 5 — When are birds most active?

The question we were asking

Does the number of species change by month, or by time of morning?

What the code did

It averaged species per session by month, then by the hour the survey started. Then it ran a statistical test to check whether the pattern was real or just random wobble.

What we found

By hour, grassland shows a clear decline: **10.06 species at 6am, dropping to 8.25 by 10am** — an 18% fall. The test confirms it is real (p is roughly 0.00005, far below the 0.05 threshold). Forest stays flat and its test comes back **not significant** ($p = 0.29$).

An everyday way to think about it. A "p-value" answers: if there were really no pattern, how likely is it we would see something this strong by chance? $p = 0.29$ means "quite likely, do not trust it". $p = 0.00005$ means "almost impossible by luck, this is real".

Why it matters

This is one of the most useful practical findings in the project: **survey grassland early**. Before 8am you find noticeably more species in the same ten minutes.

The monthly numbers are shakier. Forest had 391 sessions in June but only about 211 in May and July, so those months are not being compared on equal footing. That is why time-of-day, not month, is our headline temporal finding.

Cell 6 — Does the weather change what you find?

The question we were asking

Do temperature, humidity, sky and disturbance affect how many species get recorded?

What the code did

It tested temperature and humidity against species count, then grouped sessions into weather bands and compared the averages.

What we found

Temperature is the strongest effect and it is negative in both habitats — hotter surveys find fewer species. But the bands are revealing: richness **peaks at 15–20°C (9.61)** and falls off both above and below that. Partly cloudy skies (9.20) actually beat clear skies (8.75). Serious disturbance is devastating: **5.65 species against 8.93** when there is none.

An everyday way to think about it. It is tempting to smooth over a result that does not fit. Resist. A report that says "this was unexpected and we do not know why" is more trustworthy than one that quietly tidies it away.

Why it matters

These become real recommendations: survey in mild, partly cloudy conditions, and note disturbance because it costs you roughly 40% of what you would otherwise find.

One honest oddity. "Slight disturbance" scored *higher* (9.38) than "no disturbance" (8.93), which should not happen. Both have large samples so it is not a fluke. We do not know why. The right thing to do is report it as observed and not invent an explanation.

Cell 7 — Do the three observers count differently?

The question we were asking

Three people collected all this data. Do they record the same way? And if not, does that break our findings?

What the code did

It compared the average species per session for each observer, tested whether the differences were real, and then — crucially — checked whether each observer had worked a fair share of each habitat and park.

What we found

The differences are large and definitely real. Elizabeth averages **9.96** species per session, Brian **7.27** — a 37% gap, with statistics that leave no doubt.

But the balance check is reassuring: Forest sessions split **264 / 291 / 259** between the three, grassland **197 / 199 / 198**. Almost perfectly even.

An everyday way to think about it. Three judges score a competition, and one is consistently harsh. If every contestant is scored by all three, the harsh judge lowers everyone equally and the ranking survives. Only if he judged one group and not the other would the result be wrong.

Why it matters

At first glance a 37% observer gap looks fatal, because it is *bigger* than the habitat difference we are trying to measure.

What saves it is the balance. Bias needs **imbalance** to do damage. Because all three observers worked both habitats almost equally, their personal differences cancel out when you compare forest against grassland. They make the measurement noisier, but they do not tilt it one way.

Cell 8 — How were the birds actually detected?

The question we were asking

Were birds seen or heard? And does that differ between forest and grassland?

What the code did

It calculated what share of sightings came from singing, calling or actually seeing the bird, then looked at flyovers, distance and whether the bird's sex could be recorded.

What we found

Singing is similar in both (63.5% forest, 61.4% grassland). But **seeing** the bird is four times more common in grassland (20.1% against 5.1%), and flyovers eight times more common.

Then the striking one: sex could not be determined for **99.3% of forest sightings**, against 53.5% in grassland.

An everyday way to think about it. Trying to identify who is speaking in a crowded room versus an empty field. Same voices, very different odds of matching a face to them.

Why it matters

These two findings are actually **the same finding**. You usually cannot tell a bird's sex from its song — you have to see it. Forest visual detection is 5%, and forest sex recording is 0.7%. Grassland visual detection is 20%, and grassland sex recording is 46.5%. The numbers track each other because one causes the other.

So the empty sex column in forest is not sloppy recording. It is **a physical consequence of tree cover**. That is a far better sentence for the report, because it explains the gap instead of just noting it.

It also raises a caveat: the two habitats differ in how *detectable* birds are, not only in which birds live there.

Cell 9 — Pulling it together

The question we were asking

What did we learn, and what carries forward into the analysis?

What the code did

It printed a summary of everything above, so the findings sit in one place instead of scattered across nine outputs.

What we found

Five headings: survey effort, temporal, environmental, observer, detection. Each with the numbers that support it.

Why it matters

This block is written to be copied more or less directly into the methodology and findings sections of your report. Everything in it is traceable back to a cell you ran yourself.

What EDA changed about the rest of the project

What we found	What it forces us to do
Effort is wildly uneven across parks and habitats	Never compare raw counts. Use species per session, always.
Sessions are 10 minutes in both habitats	Per-session is a fair unit — this is what licenses the whole approach.
Seven parks have no grassland at all	Habitat comparisons only inside the four parks that have both.
Grassland richness falls through the morning	Report time-of-day as the temporal headline, not month.
Forest June sampling is double the other months	Treat monthly figures as descriptive only.
Observers differ 37% but are balanced	Disclose it, do not correct for it, and show the balance check.
Forest birds are rarely seen	Sex analysis is grassland-only, and detectability is a stated caveat.
Only 8 at-risk species, 378 sightings	Claim rates confidently, species-level patterns cautiously.

One last thing worth taking away

Notice how many of these findings are about **how the data was collected** rather than about birds. Effort, observers, detectability, sampling gaps.

That is normal, and it is the real value of EDA. Most wrong conclusions in data work do not come from bad maths. They come from someone comparing two things that were never measured the same way, and never checking. Nine cells of looking around is what stops that happening.